

Q&A

Crapware, irksome dialogs, motherboard woes, cranky GPU fans...Digit solves all your issues

Google's self-driving cars

The project is still far from being viable, but Google has set up a demonstration system on its campus.
<http://bit.ly/pEy3hE>

DIY:

Remote Missile Launcher

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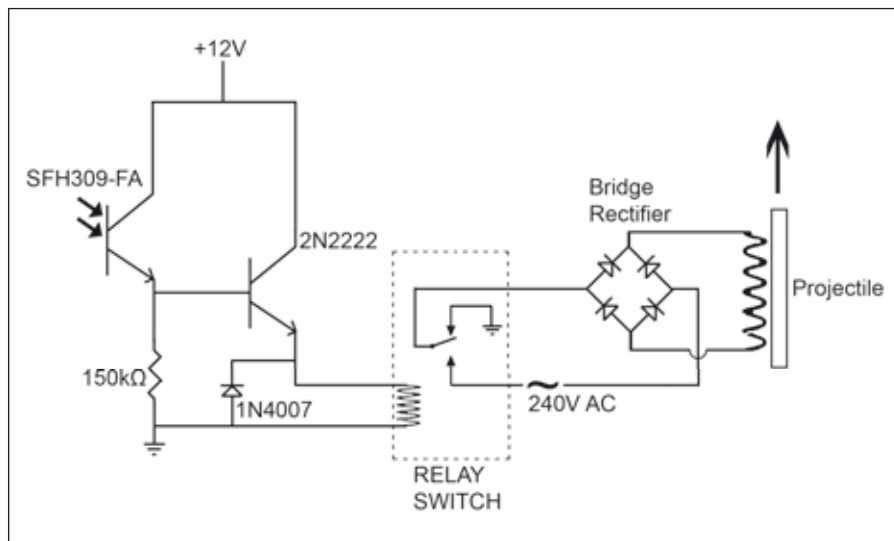
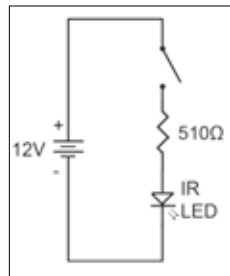
The launcher is triggered by a basic infrared remote control made using a 1.5v battery, a push-to-on switch, 510Ω resistor and a IR LED. When the switch is triggered the IR LED emits infrared light of 940nm wavelength.

The launcher consists of an air-core solenoid coil, an electromagnet which can handle 240volts AC. We sourced it from a doorbell. If it's of any help ask the guy at the hardware shop for a "ting-tong" bell. The solenoid when powered by AC voltage causes the armature to vibrate at 50-60Hz (depending on power source), so we're going to power with DC voltage. Which is in turn obtained by rectifying the voltage through a bridge rectifier. You can buy any bridge rectifier for the purpose or simply make one like we did. We rigged 4 1N4007 diodes to make the rectifier. A word of warning though: we're on a single-phase connection, a three-phase connection requires a different rectifier all together. The rectifier is in series with the AC voltage source and a relay switch. The relay switch is toggled by the phototransistor when infrared light of a specific wavelength is

focused onto it. In this case it's within the range of 380nm to 1180nm (SFH309). If you use the SFH309-FA, ensure that the IR LED is capable of emitting infrared light of 880nm wavelength or in its proximity. The phototransistor, when activated, passes 12 volts into the relay coil. This in turn toggles the relay and completes the circuit.

The projectile should be thin enough to pass through the core of the coil. Having a magnetised projectile helps as a normal ferromagnetic projectile would just get attracted to the magnet and get stuck in the core. A way around this would be to use a IC555 timer to generate a small-timed pulse. The projectile should be placed in the core of the coil and the pulse will switch the relay for a fraction of a second which makes

the coil attract the projectile towards its centre. This time period will provide initial velocity for the projectile and then before the coil starts pulling the magnet back towards the centre, it will shut down. This action will provide the projectile with an impulse force enough to fire it.



Stuff you'll need

Remote	
Component	Attributes
Switch	Push-to-on
Resistor	510Ω
IR LED	OFL-3102/OFL-5102
Battery	1.5V AA Dry Cell

Launcher	
Component	Attributes
Power Source	12V
IR Receiver	SFH-309FA-4/BPW85C/TEFT4300
Transistor	2N2222
Resistor	150KΩ
Solenoid Coil	Sourced from calling bell
Magnetic armature	Sourced from calling bell and modified
Diode	5 x 1N4007
Relay	Coil voltage - 12V DC, Contact voltage - 240V AC
Connecting wires	As needed

Now, you can set up the launcher to knock your boss' door every two minutes to pester him. Or you could tie a message around the projectile and shoot a message across balconies. Use your imagination! Have fun and write in with your queries. [d](#)

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